

Alex Gold

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EDUCATION

Brigham Young University

Expected Apr 2024

Bachelor of Science: Mechanical Engineering

Provo, UT

- Major GPA 3.78
- Relevant Coursework: Power Electronics, Heat Transfer, Materials Engineering, MATLAB, Diff. Eq., CAD, Thermodynamics, Fluids, Mechatronic Systems

ENGINEERING EXPERIENCE

Lithium-Ion Battery Lab at BYU

September 2022-Current

Research Assistant- Dr. Dean Wheeler & Dr. Brian Mazzeo

Provo, UT

- Assisted and took the lead in experiments over the course of 3 semesters, focused on manufacturing experiment hardware and refining existing processes
- Constructed single layer pouch batteries using experimental silicon samples and ran charge/discharge tests
- Built high pressure test cells for lithium-oxygen battery experiments and ran discharge tests
- Instructed to find a way to turn a key reduction reaction into a continuous flow process as part of a silicon refinement procedure
- Failed several times while trying to build a high heat and pressure (250 C, 5 bar) system to meet this goal, designed and manually machined several parts, carried out extensive testing
- Worked with professors on a second version using a peristaltic pump and software-controlled valves

BYU Racing Formula SAE EV

July 2023-Current

Capstone team: Accumulator

Provo, UT

- Designed a custom 450v, 8kWh battery pack for a first-year formula E competition car
- Led a team of 9 engineers in the design of the cell submodule, collection plates, BMS connection, and custom integrated fuses
- Wrote MATLAB scripts and used thermal FEA to choose fuse design
- Assisted in designing the cooling system, precharge/discharge circuit, shutdown circuit, and enclosure.
- Produced rigorous documentation of design choices and justifications

High Voltage Sub-team

August 2022-July 2023

- Conducted initial research and planning as founding member of team
- Researched and built high voltage battery pack from scratch and integrated it into a 20kW drivetrain
- Used decision matrices to optimize hardware and software choices
- Coordinated with overseas suppliers to procure parts

Lion Energy

May 2023-September 2023

Mechanical Engineering Intern

American Fork, UT

- Designed, modeled in CAD, and built the first prototype of a 4.5 kWh smart modular battery pack
- Ran thermal analysis of the battery layout in SolidWorks FEA to determine active cooling requirements
- Ran static analysis of the container and rack in SolidWorks FEA to size the internal supports and rack
- Generated engineering drawings and sourced parts from 11 third party manufacturers
- Actively participated in design reviews with electrical, software, validation, and management teams

International Energy Study Abroad

April-May 2022

Participant

Italy, Denmark, Sweden, Utah

- Investigated 24 international energy sites including nuclear, geothermal, NGCC, coal, wind, and hydro power plants, as well as manufacturing and testing facilities for wind and solar

- Gained a deep understanding of the global energy market, as well as the opportunities and issues with renewables
- Researched lithium-ion batteries as a grid storage solution and compiled findings in a paper

ENGINEERING PROJECTS

Personal Projects and Ventures

2016-Current

- 2017-2023: Designed and sewed 6 outdoor-use bags
- 2020: Designed and prototyped a cost-efficient backcountry skiing system
- 2022: Designed in CAD and used CNC to prototype custom dish rack
- 2022: Designed and built a single-sensor frameless laser harp using Arduino and C language

Clubs

- BYU Racing (Formula E SAE)
- Engineering Design Association (as marketing director)
- Mechatronics Club

SKILLS/INTERESTS

- Talents and Interests: skiing, hiking, film photography, prototyping, and creative design
- Skills: SolidWorks, MATLAB, C++, Ansys, Manual Machining, LT SPICE, Microcontrollers, Power electronics